A PENCIL ON A NONSINGULAR PROPER CURVE GIVES EITHER A MAP OF DEGREE AT MOST D OR A TRIVIAL MAP WITH NONTRIVIAL GLOBAL FUNCTIONS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the degree bounds, the trivial case with the H^0 consequence, and the nonconstant case with the degree formula are all retained verbatim.

Editorial note. The label lemma-g1d names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The morphism attached to a pencil. Same chapter, Lemma lemma-linear-series. Produces the morphism phi: X -> P^1 of the g^1_d with L' = phi^*O(1) receiving a nonzero map from the pencil.
- Degree bookkeeping for invertible sheaves. Varieties, Lemmas varieties-lemma-check-invertible-sheaf-trivial and varieties-lemma-degree-pullback-map-proper-curves. Bounds 0 <= deg(L') <= d, forces L' trivial at degree zero, and gives the degree formula for the pullback under a nonconstant map.

Source reference. Original reference 0CCP: sources/stacks-curves/pinned-original-source.tex, source lines 583–583. Prerequisite pointer invoked at proof line(s) 629; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0B40: sources/stacks-chow/tags.txt, source lines 13983–13983. Prerequisite pointer invoked at proof line(s) 642; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AYZ: sources/stacks-chow/tags.txt, source lines 13807–13807. Prerequisite pointer invoked at proof line(s) 654; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a nonsingular proper curve over k. Let $(\mathcal{L}, V)$ be a $\mathfrak{g}^1_d$ on X. Then the morphism $\varphi : X \to \mathbf{P}^1_k$ of Lemma 0CCP either*

(1) *is nonconstant and has degree $\leq d$, or*
(2) *factors through a closed point of $\mathbf{P}^1_k$ and in this case $H^0(X, \mathcal{O}_X) \neq k$.*

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Proof. By Lemma 0CCP we see that $\mathcal{L}' = \varphi^*\mathcal{O}_{\mathbf{P}^1_k}(1)$ has a nonzero map $\mathcal{L}' \to \mathcal{L}$. Hence by Varieties, Lemma 0B40 we see that $0 \leq \deg(\mathcal{L}') \leq d$. If $\deg(\mathcal{L}') = 0$, then the same lemma tells us $\mathcal{L}' \cong \mathcal{O}_X$ and since we have two linearly independent sections we find we are in case (2). If $\deg(\mathcal{L}') > 0$ then φ is nonconstant (since the pullback of an invertible module by a constant morphism is trivial). Hence

$$\deg(\mathcal{L}') = \deg(X/\mathbf{P}^1_k)\deg(\mathcal{O}_{\mathbf{P}^1_k}(1))$$

by Varieties, Lemma 0AYZ. This finishes the proof as the degree of $\mathcal{O}_{\mathbf{P}^1_k}(1)$ is 1. $\square$